MATH3018 Assignment 1, due on Jan 22

(1) Given the following data matrix $\mathbf{X}$ with 4 observations and 2 variables

$$\mathbf{X} = \begin{pmatrix} 1 & 3 \\ 7 & 6 \\ 5 & 4 \\ 7 & 3 \end{pmatrix}$$

Find the following:
(a) Sample mean vector $\overline{\mathbf{x}}$. (1pt)
(b) Sample covariance matrix $\mathbf{S}$. (2pts)
(c) $\mathbf{S}^{-1}$. (1pt)
(d) Sample correlation matrix $\mathbf{R}$. (1pt)

Solution

(a) $\overline{\mathbf{x}} = \frac{1}{n}\mathbf{X}^T\mathbf{1}_n = (5,4)^T$.

(b)

$$\mathbf{S} = \frac{1}{n-1}\left(\mathbf{X}^T\mathbf{X} - n\overline{\mathbf{x}}\overline{\mathbf{x}}^T\right) = \begin{pmatrix} 8 & 2 \\ 2 & 2 \end{pmatrix}$$

(c)

$$\mathbf{S}^{-1} = \frac{1}{12}\begin{pmatrix} 2 & -2 \\ -2 & 8 \end{pmatrix}$$

(d) Let

$$\mathbf{D} = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}.$$

(2) Prove the following statements: (6pts)

(a) The determinant of a $p \times p$ orthogonal matrix $\mathbf{Q}$ is either 1 or -1. (can use properties listed on page 24 of Chapter 2 slides directly)

(b) $\mathbf{A}$ is a symmetric matrix, and λ_i and λ_j are two distinct eigenvalues of $\mathbf{A}$, then the corresponding eigenvectors $\mathbf{u}_i$ and $\mathbf{u}_j$ must be orthogonal, i.e. $\mathbf{u}_i^T\mathbf{u}_j = 0$. (Hint: start from the definition of eigenvalue/eigenvector)

(c) For any matrix $\mathbf{A}$, $\mathbf{A}^\top\mathbf{A}$ is positive semi-definite.

(d) Sample covariance matrix is positive semi-definite.

(e) For any $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{n\times m}$, $tr(\mathbf{AB}) = tr(\mathbf{BA})$.

(f) The trace of a symmetric $p \times p$ matrix $\mathbf{A}$ can be expressed as the sum of its eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_p$.

Solution

(a) $QQ^T = I_p$. Thus $1 = |I_p| = \left|QQ^T\right| = |Q|\left|Q^T\right| = |Q|^2$, which implies that $|Q| = 1$ or -1 .

(b) $A\mathbf{u}_i = \lambda_i \mathbf{u}_i$ implies $\mathbf{u}_j^T A \mathbf{u}_i = \mathbf{u}_j^T \lambda_i \mathbf{u}_i = \lambda_i \mathbf{u}_j^T \mathbf{u}_i$; $A\mathbf{u}_j = \lambda_j \mathbf{u}_j$ implies $\mathbf{u}_i^T A \mathbf{u}_j = \mathbf{u}_i^T \lambda_j \mathbf{u}_j = \lambda_j \mathbf{u}_i^T \mathbf{u}_j$. Since $\mathbf{u}_j^T A \mathbf{u}_i = \mathbf{u}_i^T A \mathbf{u}_j$, we have $\lambda_i \mathbf{u}_j^T \mathbf{u}_i = \lambda_j \mathbf{u}_i^T \mathbf{u}_j$. And by $\lambda_i \neq \lambda_j, \mathbf{u}_i^T \mathbf{u}_j = 0$.

(c) For any vector $\mathbf{v} \in \mathbb{R}^p$, $\mathbf{v}^\top \mathbf{A}^\top \mathbf{A} \mathbf{v} = (\mathbf{A}\mathbf{v})^\top (\mathbf{A}\mathbf{v}) \geq 0$. Thus $\mathbf{A}^\top \mathbf{A}$ is positive semi-definite.

(d) Let $\mathbf{y}_i$ be the ith centered observation, i.e., $\mathbf{y}_i = \mathbf{x}_i - \bar{\mathbf{x}}$. Then the sample covariance matrix can be written as $\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n \mathbf{y}_i \mathbf{y}_i^\top$

For any vector $\mathbf{v}$,

$$
\begin{aligned}
\mathbf{v}^\top \mathbf{S} \mathbf{v} &= \mathbf{v}^\top \left(\frac{1}{n-1} \sum_{i=1}^n \mathbf{y}_i \mathbf{y}_i^\top \right) \mathbf{v} \\
&= \frac{1}{n-1} \sum_{i=1}^n \mathbf{v}^\top \mathbf{y}_i \mathbf{y}_i^\top \mathbf{v} \\
&= \frac{1}{n-1} \sum_{i=1}^n (\mathbf{y}_i^\top \mathbf{v})^2 \\
&\geq 0
\end{aligned}
$$

Therefore $\mathbf{S}$ is positive semi-definite.

(e) Let a_{ij} be elements of $\mathbf{A}$ and b_{ij} be elements of $\mathbf{B}$. We have

$$
tr(\mathbf{AB}) = \sum_{i=1}^m (\mathbf{AB})_{ii} = \sum_{i=1}^m \sum_{k=1}^n a_{ik} b_{ki}
$$

Similarly,

$$
tr(\mathbf{BA}) = \sum_{j=1}^n (\mathbf{BA})_{jj} = \sum_{j=1}^n \sum_{k=1}^m b_{jk} a_{kj} = \sum_{k=1}^m \sum_{j=1}^n a_{kj} b_{jk} = \sum_{i=1}^m \sum_{k=1}^n a_{ik} b_{ki}
$$

Thus we conclude the proof.

(f) $\operatorname{tr}(A) = \operatorname{tr}\left(U \Lambda U^T\right) = \operatorname{tr}\left(\Lambda U^T U\right) = \operatorname{tr}(\Lambda) = \lambda_1 + \lambda_2 + \cdots + \lambda_p$.

(3)

$$
A = \begin{pmatrix} 4 & -\sqrt{2} \\ -\sqrt{2} & 3 \end{pmatrix}
$$

(a) Calculate the eigenvalues λ_1 and λ_2 as well as the eigenvectors $\mathbf{u}_1$ and $\mathbf{u}_2$ of A (ensure that $\mathbf{u}_1$ and $\mathbf{u}_2$ both of unit length). Write down the eigenvector matrix U (of A) and calculate the inverse U^{-1} using the property of orthogonality. (2pts)

(b) Use the eigenvalue decomposition (EVD) to find A^5. (1pt)

(c) Use the EVD to find $A^{1/2}$. (1pt)

Solution

(a) $\lambda_1 = 5, \lambda_2 = 2$. $\mathbf{u}_1 = (\sqrt{6}/3, -\sqrt{3}/3)^T, \mathbf{u}_2 = (\sqrt{3}/3, \sqrt{6}/3)^T$.

$$U = [\mathbf{u}_1, \mathbf{u}_2] = \frac{1}{3}\begin{pmatrix} \sqrt{6} & \sqrt{3} \\ -\sqrt{3} & \sqrt{6} \end{pmatrix}, U^{-1} = U^T = \frac{1}{3}\begin{pmatrix} \sqrt{6} & -\sqrt{3} \\ \sqrt{3} & \sqrt{6} \end{pmatrix}$$

(b)

$$A^5 = U\Lambda^5 U^T = \begin{pmatrix} 2094 & -1031\sqrt{2} \\ -1031\sqrt{2} & 1063 \end{pmatrix}$$

(c)

$$A^{1/2} = U\Lambda^{1/2} U^T = \frac{1}{3}\begin{pmatrix} 2\sqrt{5}+\sqrt{2} & 2-\sqrt{10} \\ 2-\sqrt{10} & \sqrt{5}+2\sqrt{2} \end{pmatrix}$$